

EXPERIMENT – 13

Study of Phishing Attacks Using Z-Phisher (Open-Source GitHub Tool) in Kali Linux

Aim:

To study phishing attack concepts using the Z-Phisher open-source framework in Kali Linux in order to understand social-engineering threats, user behavior exploitation, and preventive security measures.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Study Tool:** Z-Phisher (Open-Source GitHub Framework – Educational Use)
- **System Requirements:**
 - RAM: Minimum 4 GB (8 GB recommended)
 - Disk Space: Minimum 30 GB
 - Processor: Intel/AMD with Virtualization support

Procedure:

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

Step 2: Study the Z-Phisher Framework

1. Visit the Z-Phisher GitHub repository (read-only study).
2. Review documentation to understand:
 - Types of phishing templates
 - Social-engineering techniques used
 - Ethical usage warnings

Step 3: Understand Phishing Attack Workflow

1. Analyze how fake login pages mimic legitimate websites.
2. Observe how users are tricked into entering credentials.
3. Study the role of URL masking and trust exploitation.

Step 4: Controlled Demonstration (Simulation Only)

1. Perform a **local, consent-based simulation** using dummy accounts.
2. Observe how phishing pages appear to users.
3. Analyze user response behavior (no real credentials used).

Step 5: Identify Security Risks

- Credential theft
- Identity compromise
- Financial fraud
- Privacy violations

Step 6: Study Prevention Techniques

- User awareness and phishing education
- Browser security warnings
- Two-factor authentication (2FA)
- Email and URL filtering
- HTTPS and domain verification

Output:

```
(kali㉿kali)-[~/zphisher]
$ git clone https://github.com/htr-tech/zphisher.git
Cloning into 'zphisher' ...
remote: Enumerating objects: 1801, done.
remote: Total 1801 (delta 0), reused 0 (delta 0), pack-reused 1801 (from 1)
Receiving objects: 100% (1801/1801), 28.68 MiB | 13.82 MiB/s, done.
Resolving deltas: 100% (817/817), done.
```

```
(kali㉿kali)-[~/zphisher]
$ cd zphisher
```

```
(kali@kali)-[~/zphisher/zphisher]
$ bash zphisher.sh
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  htr-tech
  Version : 2.3.5

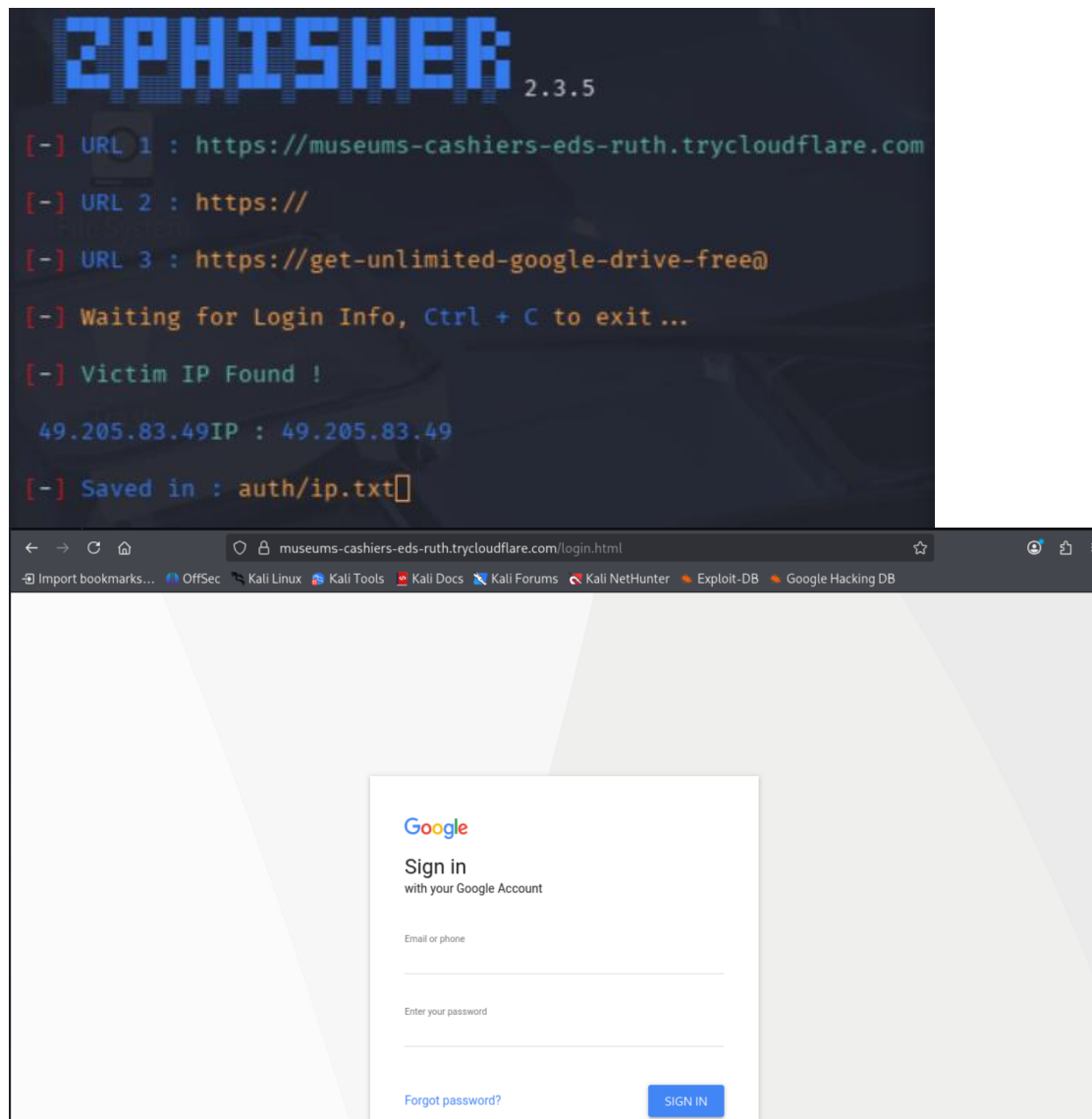
[-] Tool Created by htr-tech (tahmid.rayat)

[::] Select An Attack For Your Victim [::]

[01] Facebook          [11] Twitch              [21] DeviantArt
[02] Instagram         [12] Pinterest           [22] Badoo
[03] Google            [13] Snapchat            [23] Origin
[04] Microsoft         [14] LinkedIn            [24] DropBox
[05] Netflix           [15] Ebay               [25] Yahoo
[06] Paypal            [16] Quora              [26] Wordpress
[07] Steam             [17] Protonmail          [27] Yandex
[08] Twitter           [18] Spotify            [28] StackoverFlow
[09] Playstation       [19] Reddit             [29] Vk
[10] Tiktok            [20] Adobe               [30] XBOX
[31] Mediafire         [32] Gitlab              [33] Github
[34] Discord           [35] Roblox

[99] About             [00] Exit

```



Result:

The study of phishing attacks using the Z-Phisher framework was successfully completed in Kali Linux.

This experiment improved understanding of social-engineering threats and emphasized the importance of user awareness and defensive security practices.